

LoRA-Adapted RETFound with Multi-Scale Fusion and Ordinal Losses for Five-Class Diabetic Retinopathy Grading and External Validation

Nilotpal Bose

Department of Computer Science and Engineering

[update-email]

Abstract

Adapting retinal foundation models for ordered DR severity remains challenging when flat cross-entropy ignores class imbalance and clinical ordinal structure. We compare RETFound Baseline (full fine-tuning with weighted cross-entropy) and Enhanced RETFound (LoRA adapters, multi-scale token fusion, and focal/ordinal/referable losses) for five-class diabetic retinopathy grading. On APTOS 2019 (test N = 550), Enhanced RETFound slightly improved quadratic weighted kappa (QWK 0.893 vs 0.888) and referable AUROC (0.987 vs 0.981), while the baseline retained higher accuracy (0.831 vs 0.811). On external Messidor-2 without fine-tuning, Enhanced RETFound improved QWK (0.505 vs 0.488) and referable AUROC (0.778 vs 0.725). Ablations show A2 (LoRA + multi-scale + focal) achieves the best APTOS QWK (0.895), whereas ordinal/referable auxiliaries mainly boost screening AUROC. Overall, Enhanced RETFound is comparable in-domain and stronger under external shift for ordinal and screening-oriented metrics.

Index Terms: diabetic retinopathy, RETFound, foundation model, LoRA, ordinal regression, QWK, external validation.

1 Introduction

Diabetic retinopathy is a microvascular complication of diabetes and a major cause of preventable vision loss worldwide. Population screening relies on grading colour fundus photographs according to clinical severity scales such as ICDR (grade 0 to grade 4). Because grading is labour-intensive and subject to inter-observer variability, deep learning has been widely explored for automated DR detection and staging.

Early systems were typically trained from scratch or from ImageNet-initialised convolutional networks on large labelled fundus corpora. More recently, foundation models pretrained with self-supervised learning on massive unlabelled medical image collections have emerged as a stronger starting point for downstream clinical tasks. RETFound is a retinal foundation model based on a Vision Transformer pretrained with masked autoencoding on approximately 1.6 million fundus and OCT images.

Despite this progress, adapting RETFound to DR severity grading still raises open questions. Standard full fine-tuning with flat cross-entropy treats grades as unordered classes and

may under-emphasise rare mild/severe cases and ordinal structure. Parameter-efficient fine-tuning methods such as LoRA, multi-scale feature aggregation, and imbalance-/ordinal-aware losses offer a more clinically aligned adaptation recipe.

2 Materials and Methods

2.1 Dataset Description

APTOS 2019 Blindness Detection provides labelled macular-centred fundus photographs with ICDR-aligned grades 0–4. We used a stratified 70/15/15 train/validation/test split fixed in `splits.json` (test N = 550). Images were resized to 224×224 , normalised with ImageNet statistics, and lightly augmented.

Messidor-2 is used strictly as an external test set (no fine-tuning). We report both five-class grading and referable DR (grade ≥ 2).

2.2 Frameworks for Comparison

Two frameworks share the same RETFound ViT-Large/16 colour-fundus pretrained backbone and the same preprocessing.

- **RETFound Baseline:** full encoder fine-tuning + weighted cross-entropy.
- **Enhanced RETFound:** frozen backbone with LoRA adapters on attention `qkv` (1.3% trainable parameters), multi-depth feature fusion (blocks 7, 15, 23), and focal + CORAL ordinal + referable auxiliary losses.

We also evaluate three 15-epoch ablations: A1 (LoRA + late + focal), A2 (A1 + multi-scale), A3 (A2 + ordinal/referable auxiliaries).

2.3 Metrics

We report accuracy, macro-F1, QWK, referable accuracy/AUROC, confusion matrices, and ROC curves. Primary interpretation emphasises QWK and referable AUROC.

Table 1: Main results (checkpoints selected by best validation QWK).

Model	Set	Acc	M-F1	QWK	R-Acc	R-AUC
Baseline	APTOS	0.831	0.671	0.888	0.929	0.981
Enhanced	APTOS	0.811	0.683	0.893	0.935	0.987
Baseline	Messidor-2	0.612	0.330	0.488	0.807	0.725
Enhanced	Messidor-2	0.593	0.349	0.505	0.802	0.778

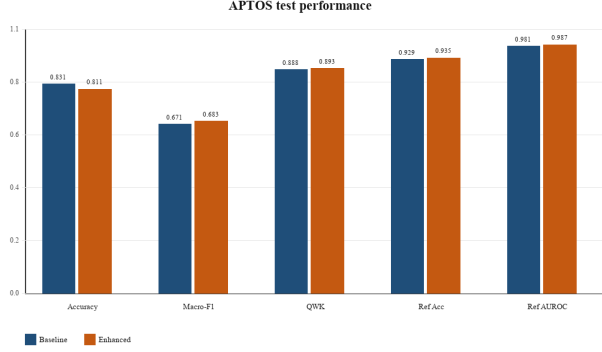


Figure 1: APTOS test metrics for RETFound Baseline versus Enhanced RETFound.

3 Results

3.1 Main Comparison on APTOS and Messidor-2

Enhanced RETFound yields a small in-domain gain in QWK (+0.005) and referable AUROC (+0.006), while Baseline remains higher in overall accuracy (+0.020). Externally, Enhanced improves QWK (+0.017) and referable AUROC (+0.053), indicating stronger robustness for clinically relevant endpoints.

3.2 Validation QWK Curves

3.3 External Validation and Error Structure

3.4 APTOS Confusion Matrices and Ablations

4 Discussion

The results suggest metric-dependent trade-offs. For strict five-class ordinal agreement, A2 (LoRA + multi-scale + focal) performs best. For screening-focused use, auxiliary objectives improve referable AUROC, especially under external shift.

5 Conclusion

Enhanced RETFound is comparable to full fine-tuning in-domain and stronger externally on ordinal/screening metrics. Foundation-model adaptation for DR should be assessed with clinically meaningful endpoints and external validation, not accuracy alone.

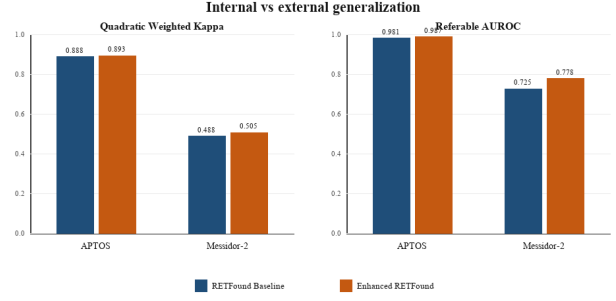


Figure 2: Internal (APTOS) vs external (Messidor-2) QWK and referable AUROC.

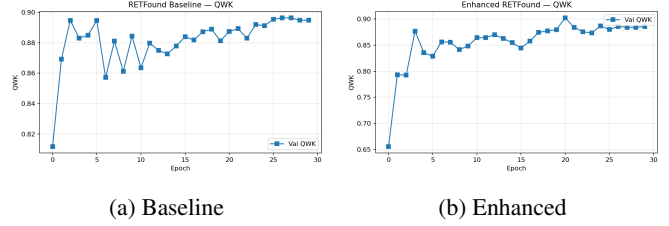


Figure 3: Validation QWK for main models.

References

1. Y. Zhou et al., “A foundation model for generalizable disease detection from retinal images,” *Nature*, 2023.
2. A. Dosovitskiy et al., “An image is worth 16x16 words,” *ICLR*, 2021.
3. K. He et al., “Masked autoencoders are scalable vision learners,” *CVPR*, 2022.
4. E. J. Hu et al., “LoRA: Low-rank adaptation of large language models,” *ICLR*, 2022.
5. T.-Y. Lin et al., “Focal loss for dense object detection,” *ICCV*, 2017.
6. W. Cao et al., “Rank consistent ordinal regression for neural networks,” *Pattern Recognit. Lett.*, 2020.
7. C. P. Wilkinson et al., “Proposed international clinical DR severity scales,” *Ophthalmology*, 2003.
8. V. Gulshan et al., “Development and validation of a deep learning algorithm for DR detection,” *JAMA*, 2016.
9. E. Decencière et al., “Feedback on a publicly distributed image database: the Messidor database,” *Image Anal. Stereol.*, 2014.
10. APTOS 2019 Blindness Detection, Kaggle: <https://www.kaggle.com/competitions/aptos2019-blindness-detection>.
11. J. Krause et al., “Grader variability and reference standards for DR ML evaluation,” *Ophthalmology*, 2018.
12. D. S. W. Ting et al., “Artificial intelligence and deep learning in ophthalmology,” *Br. J. Ophthalmol.*, 2019.

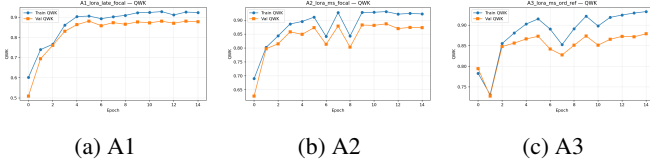


Figure 4: Validation QWK for ablations (15 epochs).

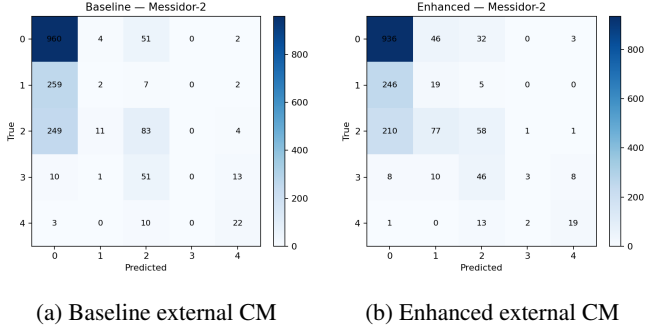


Figure 5: Messidor-2 confusion matrices.

13. T. Li et al., “Diagnostic assessment of deep learning algorithms for DR screening,” *Inf. Sci.*, 2019.
14. J. Cohen, “Weighted kappa,” *Psychol. Bull.*, 1968.
15. R. Bommasani et al., “On the opportunities and risks of foundation models,” *arXiv:2108.07258*, 2021.

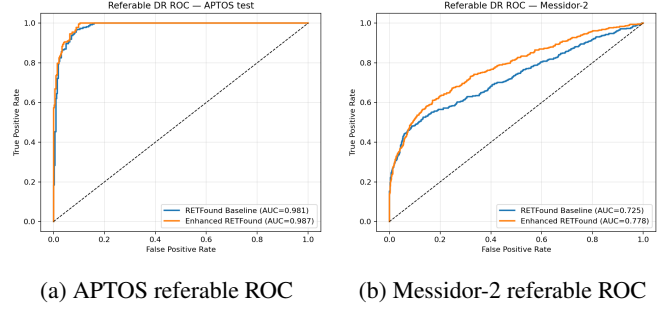


Figure 6: Referable DR ROC comparison.

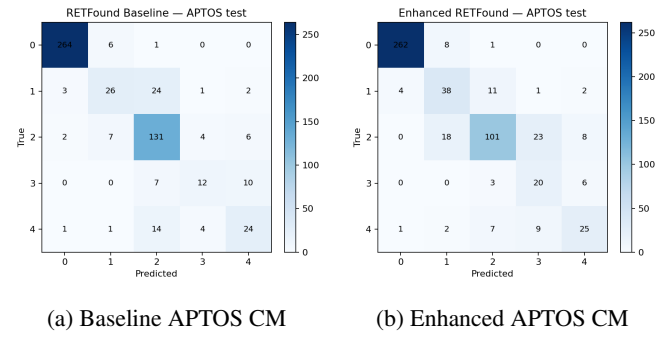


Figure 7: APTOS confusion matrices.

Table 2: Ablation results on APTOS (15 epochs; best validation-QWK checkpoint).

Variant	Acc	QWK	Ref-AUC
Baseline	0.831	0.888	0.981
Enhanced	0.811	0.893	0.987
A1	0.738	0.892	0.985
A2	0.765	0.895	0.987
A3	0.764	0.881	0.988

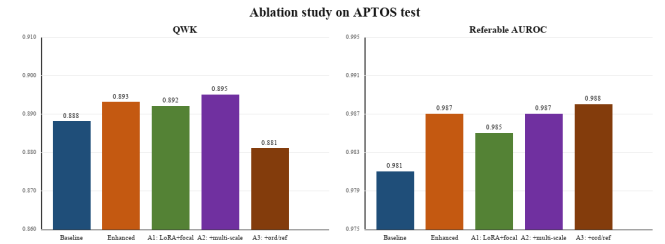


Figure 8: Ablation summary on APTOS: QWK and referable AUROC.